

Description of Preprocessing, Feature Extraction, and Model Architecture

Preprocessing

The audio recordings used in this project were provided in `.wav` format. Before training the model, several preprocessing steps were performed to ensure consistency and improve the quality of the input data.

Initially, all audio files were loaded using the Librosa library. Since audio samples may have different sampling rates, every recording was resampled to a uniform sampling frequency of **16 kHz**. This standardization ensured that all audio signals were represented consistently.

As the duration of audio recordings varied across the dataset, the extracted features were adjusted to a fixed size. Spectrograms that were shorter than the required length were padded with zeros, whereas longer spectrograms were truncated. This enabled efficient batch processing during model training.

Feature Extraction

Instead of using raw audio waveforms directly, **Mel spectrograms** were employed as feature representations. A Mel spectrogram provides a time-frequency representation of an audio signal and closely approximates the way humans perceive sound.

For each audio recording, a Mel spectrogram with **128 Mel frequency bands** was generated using the Librosa library. The power spectrogram was then converted into the **decibel (dB) scale**, which enhances the visibility of important spectral characteristics.

The resulting Mel spectrograms were resized to a fixed dimension of **128 × 100** through padding or truncation. These spectrograms were subsequently converted into PyTorch tensors and used as inputs to the classification model.

The feature extraction process can be summarized as follows:

WAV Audio File → Audio Loading → Resampling to 16 kHz → Mel Spectrogram Generation → Power-to-dB Conversion → Padding/Truncation → PyTorch Tensor Conversion.

Model Architecture

A **Convolutional Neural Network (CNN)** was used to classify speech recordings as either **Genuine (Human)** or **Deepfake (AI-Generated)**.

CNNs are particularly effective in learning spatial patterns from image-like data. Since Mel spectrograms can be treated as two-dimensional images representing the frequency content of speech over time, CNNs are well suited for this task.

The proposed model consisted of multiple convolutional layers that automatically learned important acoustic features from the Mel spectrograms. These layers were followed by activation functions and pooling operations to extract increasingly abstract representations while reducing computational complexity.

The high-level features obtained from the convolutional layers were flattened and passed through fully connected layers to perform the final classification. The output layer produced probabilities corresponding to the two classes: Genuine and Deepfake.

The model was trained using the **Cross-Entropy Loss** function and optimized using the **Adam optimizer**. Training was conducted for **five epochs**, during which both training and validation performance were monitored to ensure effective learning and to minimize overfitting.

The final trained model demonstrated excellent generalization capability and achieved high performance on previously unseen audio recordings.